# Modeling Loci and Intersections with Conic Sections

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Circles built from a distance condition, parabolas built from a focus and a directrix, intersections of a line with a conic, and intersections of two conics. Two ideas run through the whole sheet:

- **A locus is a sentence about distance.** Translate the sentence into a distance equation, square both sides, and simplify. A circle is "distance to one point is constant"; a parabola is "distance to a point equals distance to a line".
- **An intersection is a system.** Substitute one equation into the other, solve the single-variable equation that results, then substitute back for the partner coordinate. Report ordered pairs, not just $x$ values.
- Reference forms: $(x-h)^2+(y-k)^2=r^2$ for a circle with center $(h,k)$; $x^2=4py$ for a parabola with vertex at the origin, focus $(0,p)$ and directrix $y=-p$.

**1.** A circle has center $(3,-2)$ and radius 5.
Evaluate $(x-3)^2+(y+2)^2$ at the point $(7,1)$, and state whether that point lies on the circle.

Answer: ________________

**2.** A circle is the set of all points 5 units from the origin, so its equation is $x^2+y^2=25$.
Find both $y$ values on this circle when $x=3$.

Answer: ________________

**3.** A parabola has focus $(0, 3)$ and directrix $y = -3$, so $p = 3$ and its equation is $x^2 = 4py$. Find the $y$ coordinate of the point on this parabola whose $x$ coordinate is 6.

Answer: ______________

**4.** For the same parabola from problem 3, $x^2 = 12y$.
Find both $x$ values on the parabola when $y = 12$.

Answer: ______________

**5.** Find every point where the line $y = x + 1$ meets the circle $x^2 + y^2 = 25$. Give your answers as ordered pairs.

Answer: ______________

**6.** Complete the square to rewrite

$$x^2 + y^2 - 6x + 8y - 11 = 0$$

in center-radius form, then state the center and the radius.

Answer: ________________

**7.** Find every point where the circle $x^2 + y^2 = 25$ meets the parabola $y = x^2 - 5$. Give your answers as ordered pairs.

Answer: ________________

**8.** Find every point where the line $y = 2x - 3$ meets the parabola $y = x^2 - 2x + 1$, and say what the number of solutions tells you about how the line sits against the curve.

Answer: ________________

**9.** A point $P(x, y)$ moves so that it is always the same distance from the point $(0, 5)$ as it is from the line $y = -5$.

(a) Starting from $\sqrt{x^2 + (y-5)^2} = y + 5$, derive the equation of the locus and simplify it.

(b) Using your equation, find both points on the locus with $y = 5$.

(b) ________________

**10.** Find every point where the circle $x^2 + y^2 = 25$ meets the hyperbola $xy = 12$. Give your answers as ordered pairs.

Answer: ________________

**11.** A point $P(x, y)$ moves so that its distance from $A(6, 0)$ is always twice its distance from $B(0, 0)$.

Write $PA = 2 \cdot PB$ as a distance equation, square both sides, and simplify until the locus is in center-radius form. State the center and the radius.

Answer: ________________

**12. Synthesis challenge.** The cross-section of a satellite dish is the parabola $x^2 = 8y$, whose focus is $(0, 2)$ and whose directrix is $y = -2$. A circular guard ring around it has equation $x^2 + y^2 = 20$.

(a) Find every point where the ring meets the dish.

(a) ________________

(b) Take the intersection point with positive $x$. Find its distance from the focus $(0, 2)$, and separately find its distance from the directrix $y = -2$.

(b) ________________

(c) Explain what your two answers in part (b) say about every point of the dish.